

Kabilan S

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RTL Design Engineer aspirant with hands-on experience in digital systems, embedded hardware, and low-level programming. Strong foundation in Verilog, processor design, AXI protocol, and hardware debugging, with practical implementation experience in CPU architecture and embedded platforms. Skilled in digital communication protocols, including UART, SPI, I2C, and RS232. Quick learner with strong analytical and problem-solving abilities, actively seeking opportunities in RTL Design and VLSI engineering.

Technical Skills

- **RTL Design:** Verilog, System Verilog, RTL Modeling, FSM Design, Pipelined Architecture, Low-Power Design Techniques.
- **Digital Design:** Processor Design, Cache Architecture, Memory Subsystems, Timing Analysis, Clock Domain Crossing, Synchronization Techniques.
- **BUS & Communication Protocol:** AXI, AHB, APB, UART, SPI, I2C, PCIe Basics.
- **Verification & Simulation:** SystemVerilog Assertions (SVA), UVM Basics, Testbench Development, Functional Verification, Coverage Analysis, Waveform Debugging.
- **FPGA & SOC Development:** FPGA prototyping, Xilinx Vivado, ASIC Design Flow Basics, SoC Integration, Hardware Acceleration Concepts.
- **Programming & Scripting:** Verilog, System Verilog, C, C++, Python, TCL, Shell Scripting.
- **EDA Tools & Environment:** Vivado, QuestaSim/ModelSim, Synopsys VCS Basics, GTK Wave, Git, Linux.
- **Core Engineering Skills:** Problem Solving, Hardware Debugging, Performance Optimization, Analytical Thinking, Rapid Prototyping, and End-to-End Development.

Experience

JASWA VENDING PRIVATE LIMITED | CHENNAI, TN
HARDWARE ENGINEER | MARCH 2025 – MAY 2026

- **End-to-End Product Lifecycle:** Spearheaded the full-cycle development of an automated perfume vending machine, taking the project from a literal “napkin sketch” to a **production-ready functional unit in 3 months**.
- **Mechanical Optimization:** Engineered a custom 3D enclosure in **Fusion 360** that reduced material costs by **75%** and assembly complexity by **60%** through integrated snap-fit components and optimized internal airflow.
- **Communication Reliability:** Architected a robust multi-protocol communication stack (**UART, SPI, I2C**) and integrated **MQTT for IoT Connectivity**, achieving **99.8% uptime** in device-to-cloud data synchronization.
- **Firmware Architecture:** Developed modular, **non-blocking firmware for the ESP32**, allowing for simultaneous sensor polling, LED Matrix UI updates, and motor control **without system latency**.
- **DFA/DFM Implementation:** Collaborated with local vendors to ensure the mechanical design was optimized for laser cutting, **reducing potential manufacturing defects by an estimated 30%**

IIT ROORKEE | ROORKEE, UK
RESEARCH INTERN (GOVERNMENT-FUNDED R&D PROJECT) | FEBRUARY 2024 – APRIL 2024

- **Reverse Engineering & Signal Analysis:** Performed **forensic-level reverse engineering of PCBs** and mixed-signal circuits using Digital Storage Oscilloscope (DSO), multimeters, and schematic reconstruction techniques to analyze signal paths and Analog Front-End (AFE) architectures.
- **Real-Time Application Development:** Architected a **real-time telemetry and alert management application**

using Flutter, MongoDB, and Firebase, enabling live sensor visualization, automated alert triggering, and scalable user management for monitoring systems.

- **PCB Design & Sensor Integration:** Contributed to **PCB design and embedded integration workflows** for the ADXL355B MEMS accelerometer platform, focusing on reliable sensor interfacing, compact hardware layout, and system-level validation.
- **Hardware & Embedded System Development:** Engineered and validated embedded sensing platforms for **earthquake early warning applications, integrating MEMS-based sensors**, analog signal conditioning circuits, and real-time data acquisition workflows.
- **R&D Documentation & Collaboration:** Maintained structured technical documentation, **iterative design reviews, and version-controlled development workflows** using Git and Linux-based environments within a government-funded research ecosystem.

Education

MAY 2025

Bachelor of Engineering - (Electronics & Communication Engineering) | Rajalakshmi Institute of Technology | Chennai, Tamil Nadu

8.65 CGPA • First class with distinction.

Projects

FPGA-BASED ADAPTIVE SPARSE TENSOR ACCELERATOR FOR EDGE AI

FPGA, VERILOG, RTL DESIGN, XILINX VIVADO, HARDWARE ACCELERATION | JUNE 2026

- **SPARSE-AWARE COMPUTE ARCHITECTURE:** Designed an adaptive sparse tensor accelerator that dynamically selects Dense, Sparse-A, Sparse-B, or Sparse×Sparse computation paths based on runtime matrix sparsity analysis.
- **EDGE AI-ORIENTED HARDWARE OPTIMIZATION:** Leveraged sparsity-aware execution to eliminate redundant multiply-accumulate operations, reducing computational workload for sparse neural-network and tensor-processing applications.
- **SPARSE MATRIX PROCESSING ENGINE:** Implemented custom sparse encoding, coordinate matching, and Sparse×Sparse matrix multiplication modules using RTL-based MAC architectures and indexed accumulation logic.
- **CONTROL & VERIFICATION:** Developed sparsity analyzers, adaptive controllers, and computation scheduling logic; verified functionality through RTL simulation, waveform analysis, and timing validation in Xilinx Vivado.
- **PERFORMANCE IMPROVEMENT:** Achieved approximately 3× lower computation cycles than conventional single-MAC dense matrix multiplication for representative sparse matrix workloads.

FPGA-BASED IMAGE PROCESSING SYSTEM

FPGA, VERILOG, XILINX, IMAGE PROCESSING | OCTOBER 2024

- **RTL Image Processing Architecture:** Designed and implemented a **hardware-accelerated Sobel edge detection pipeline** in Verilog, enabling real-time extraction of X and Y image gradients on FPGA platforms.
- **Memory & Throughput Optimization:** Optimized on-chip memory access using line buffer architectures and parallel pixel processing techniques, **reducing processing latency by 33%** and improving data throughput efficiency.
- **FPGA Verification & Debugging:** Performed RTL simulation, waveform-level debugging, and functional validation using Xilinx toolchains to verify filter **accuracy, synchronization, and timing behavior**.
- **Hardware Acceleration & Performance Tuning:** Improved edge detection performance and processing efficiency through **pipelined datapath optimization and parallel hardware execution** strategies within a 3-month development cycle.

- **Processor Architecture Design:** Designed and implemented a **custom 16-bit RISC processor** in Verilog featuring ALU operations, register file architecture, program counter control, and instruction decoding logic.
- **Pipelined Datapath Development:** Engineered a **multi-stage pipelined datapath architecture** to improve instruction throughput and optimize execution efficiency across arithmetic and memory operations.
- **Instruction Set Implementation:** Developed a **custom instruction set architecture (ISA)** supporting arithmetic, logical, branching, and memory access instructions with efficient control signal generation.
- **Memory & Bus Integration:** Integrated instruction/data memory interfaces and designed **synchronous data transfer mechanisms** for reliable communication between **processing and memory subsystems**.
- **RTL Verification & Debugging:** Performed **functional simulation, waveform analysis, and RTL debugging** using ModelSim/GTKWave to validate processor timing, control flow, and instruction execution accuracy.

Activities

- **Entrepreneurship & Innovation:** Actively engaged in **Entrepreneurship Development**, focusing on the commercialization of IoT solutions and innovation-led business growth strategies. Regularly analyze market trends to bridge the gap between R&D and consumer-ready products.
- **Technical Communication (podcasting):** Producer and host of a personal podcast exploring emerging trends in technology and spirituality. This initiative has sharpened my ability to **distill complex technical concepts** into engaging narratives for diverse audiences.
- **Creative Arts & Exploration:** Passionate about **vocal performance** (singing) and global travel, which have fostered a high degree of cultural adaptability and **creative problem-solving**.
- **Team Dynamics (Cricket):** Competitive Cricket player; developed **strong collaborative leadership**, strategic thinking, and the ability to perform under high-pressure, time-sensitive situations.

CORE COMPETENCIES

- **RTL Design & Debugging:** Skilled in Verilog-based RTL development, FSM design, waveform analysis, and functional verification.
- **Digital System Design:** Strong understanding of computer architecture, pipelining, processor design, and hardware optimization techniques.
- **Problem Solving & Verification:** Experienced in systematic debugging, timing analysis, and hardware/software root-cause identification.
- **FPGA & Hardware Prototyping:** Hands-on experience in FPGA-based system implementation, embedded integration, and real-time processing architectures.
- **Rapid Learning & Adaptability:** Quick learner with the ability to work effectively in fast-paced research and development environments.